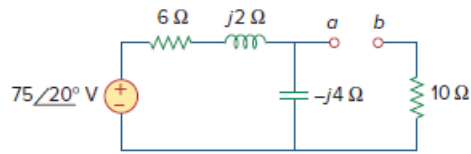


Problem

Find the Thevenin equivalent at terminals a - b of the circuit in Fig. 10.24.

Figure 10.24:



Step-by-step solution

Step 1 of 5

Refer to Figure 10.24 in the textbook.

Determine the Thevenin's impedance.

Replace the voltage source by its internal resistance. Re-draw the circuit diagram:

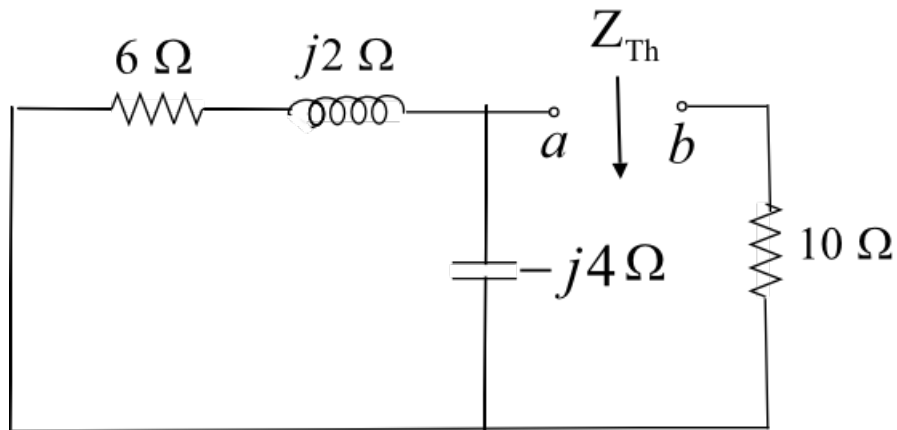


Figure 1

Step 2 of 5

The $6 + j2$ impedance is now in parallel with $-j4$ reactance. Determine the effective impedance.

$$\begin{aligned} Z_1 &= (6 + j2) \parallel (-j4) \\ &= \frac{(6 + j2)(-j4)}{6 + j2 - j4} \\ &= \frac{8 - j24}{6 - j2} \\ &= 2.4 - j3.2 \, \Omega \end{aligned}$$

Step 3 of 5

The Thevenin's impedance is the series combination of Z_1 impedance and $10\ \Omega$ resistance. Determine the Thevenin's impedance.

$$\begin{aligned} Z_{Th} &= Z_1 + 10 \\ &= 2.4 - j3.2 + 10 \\ &= 12.4 - j3.2\ \Omega \end{aligned}$$

Therefore, the Thevenin's impedance as seen into the terminals ab is $12.4 - j3.2\ \Omega$.

Step 4 of 5

Determine the value of V_{Th} , consider the circuit in figure 2.

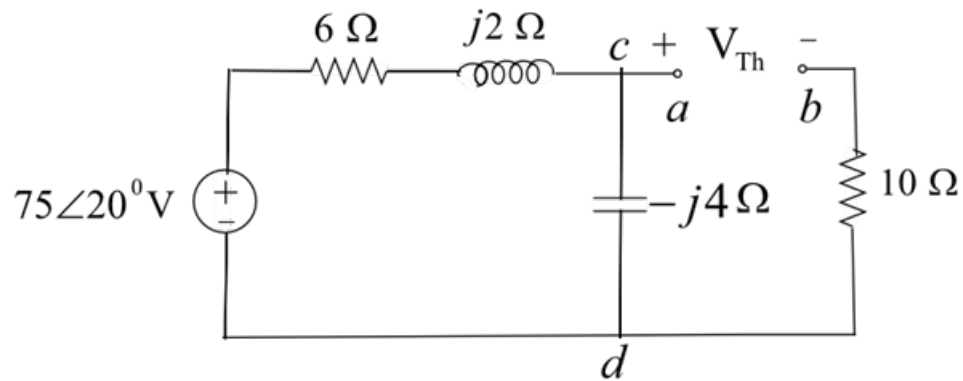


Figure 2

Step 5 of 5

The Thevenin's voltage is equal to the voltage across the terminals ab.

From Figure 2, the current passing through the $10\ \Omega$ resistance is zero.

Hence, $V_{Th} = V_{cd}$.

Apply voltage division rule and determine the value of V_{Th} .

$$\begin{aligned} V_{Th} &= 75\angle 20^\circ \times \frac{-j4}{6 + j2 - j4} \\ &= 75\angle 20^\circ \times \frac{-j4}{6 - j2} \\ &= 75\angle 20^\circ \times \frac{4\angle -90^\circ}{6.325\angle -18.43^\circ} \\ &= 47.43\angle -51.57^\circ\text{ V} \end{aligned}$$

Therefore, the Thevenin's voltage V_{Th} is $47.43\angle -51.57^\circ\text{ V}$.